

Lab-9 delivery

serializers.py with CourseSerializer.

```
class CourseSerializer(serializers.ModelSerializer):
    lessons = LessonSerializer(many=True, read_only=True)
    class Meta:
        model = Course
        fields = ['id', 'title', 'descriptions', 'durations', 'lessons']
```

serializers.py with LessonSerializer.

```
class LessonSerializer(serializers.ModelSerializer):
    class Meta:
        model = Lesson
        fields = ['id', 'title', 'content', 'video_URL']
```

API views in views.py.

```
from rest_framework.views import APIView
from rest_framework.response import Response
from rest_framework import status
from .models import Course
from .serializers import CourseSerializer

class CourseListAPI(APIView):
    def get(self, request):
        courses = Course.objects.all()
        serializer = CourseSerializer(courses, many=True)
        return Response(serializer.data)

class CourseDetailAPI(APIView):
    def get(self, request, pk):
        try:
            course = Course.objects.get(pk=pk)
        except Course.DoesNotExist:
            return Response({'error': 'Course not found'}, status=status.HTTP_404_NOT_FOUND)

        serializer = CourseSerializer(course)
```

```
return Response(serializer.data)
```

API endpoints in urls.py.

```
path('api/courses/', CourseListAPI.as_view(), name='api_course_list'),  
path('api/courses/<int:pk>', CourseDetailAPI.as_view(), name='api_course_detail'),
```

Screenshots of:

1. API response for listing courses.

The screenshot shows a REST client interface with the URL `http://127.0.0.1:8000/api/courses` and a GET request. The response is a 200 OK status with a response time of 21 ms and a size of 11.58 KB. The response body is displayed in JSON format, showing a list of three courses: Python, JavaScript, and Bio-Informatics. Each course object contains fields for id, title, descriptions, durations, lessons, and video_URL.

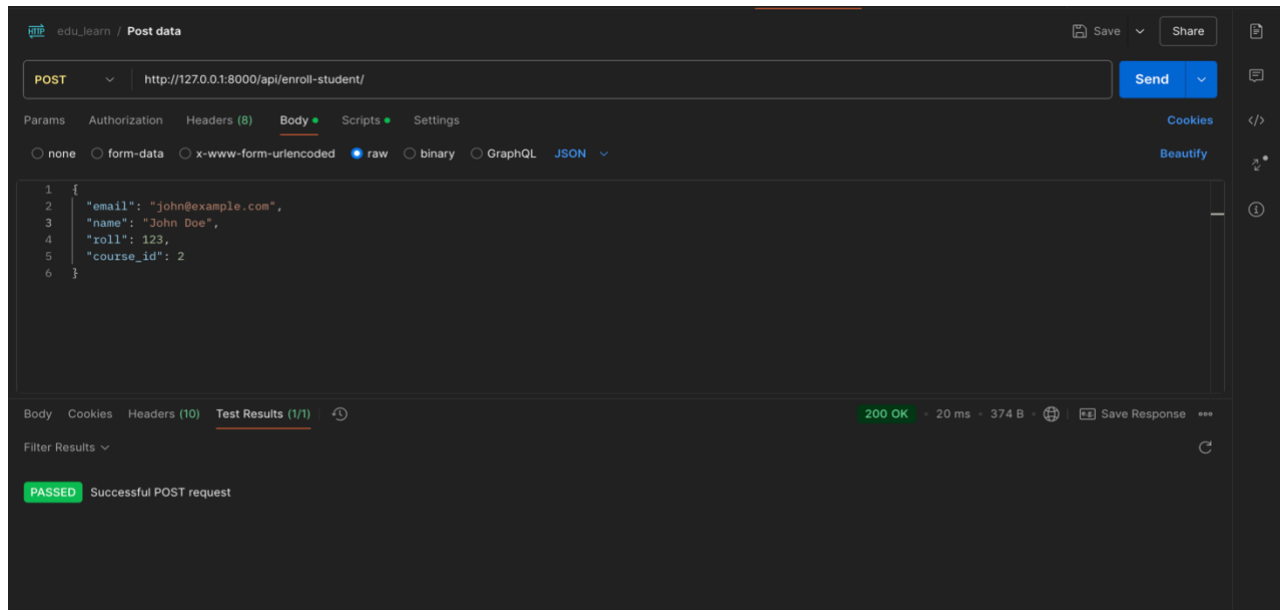
```
{  
  "id": 3,  
  "title": "Python",  
  "descriptions": "Python is a popular programming language. It was created by Guido van Rossum, and released in 1991.\r\nIt is used for: \r\nweb development (server-side),\r\nsoftware development,\r\nmathematics,\r\nsystem scripting.",  
  "durations": 40,  
  "lessons": [  
    {  
      "id": 2,  
      "title": "Python",  
      "content": "The study plan features are made to help teachers and students. They make learning easy, flexible, and fun. These features work for different types of classes, learning styles and student level.",  
      "video_URL": null  
    }  
  ],  
},  
{  
  "id": 4,  
  "title": "JavaScript",  
  "descriptions": "JavaScript is the world's most popular programming language.\r\n\r\nJavaScript is the programming language of the Web.\r\n\r\nJavaScript is easy to learn.\r\n\r\nThis tutorial will teach you JavaScript from basic to advanced.",  
  "durations": 38,  
  "lessons": [  
    {  
      "id": 3,  
      "title": "JavaScript",  
      "content": "The study plan features are made to help teachers and students. They make learning easy, flexible, and fun. These features work for different types of classes, learning styles and student level.",  
      "video_URL": null  
    }  
  ],  
},  
{  
  "id": 11,  
  "title": "Bio-Informatics",  
  "descriptions": "Bioinformatics is an interdisciplinary field that combines biology, computer science, and mathematics to analyze and interpret biological data, especially genomic and proteomic sequences",  
  "durations": 45,  
  "lessons": [  
    {  
      "id": 12,  
      "title": "Chemistry",  
      "content": "Chemistry is the science that deals with the properties, composition, and structure of substances (defined as elements and compounds), the transformations they undergo, and the energy that is released or absorbed during these processes.",  
      "video_URL": null  
    }  
  ],  
},  
{  
  "id": 19,  
  "title": "The scope of chemistry",  
  "content": "The scope of chemistry",  
  "video_URL": null  
},  
]
```

2. API response for course details.

The screenshot shows a REST client interface with the URL `http://127.0.0.1:8000/api/courses/12` and a GET request. The response is a 200 OK status with a response time of 19 ms and a size of 4.07 KB. The response body is displayed in JSON format, showing the details of the course with id 12, titled "Chemistry". The response includes fields for id, title, descriptions, durations, lessons, and video_URL.

```
{  
  "id": 12,  
  "title": "Chemistry",  
  "descriptions": "chemistry, the science that deals with the properties, composition, and structure of substances (defined as elements and compounds), the transformations they undergo, and the energy that is released or absorbed during these processes.",  
  "durations": 49,  
  "lessons": [  
    {  
      "id": 18,  
      "title": "Introduction",  
      "content": "Chemistry is the science that deals with the properties, composition, and structure of substances (defined as elements and compounds), the transformations they undergo, and the energy that is released or absorbed during these processes. \r\n\r\nEvery substance, whether naturally occurring or artificially produced, consists of one or more of the hundred-odd species of atoms that have been identified as elements. Although these atoms, in turn, are composed of more elementary particles, they are the basic building blocks of chemical substances; there is no quantity of oxygen, mercury, or gold, for example, smaller than an atom of that substance. Chemistry, therefore, is concerned not with the subatomic domain but with the properties of atoms and the laws governing their combinations and how the knowledge of these properties can be used to achieve specific purposes.",  
      "video_URL": null  
    }  
  ],  
},  
{  
  "id": 19,  
  "title": "The scope of chemistry",  
  "content": "The scope of chemistry",  
  "video_URL": null  
},  
]
```

3. Enrollment API request and response.



Lab-10 delivery

1. A deployed application URL <https://edu-learn.up.railway.app/>
2. Code for:
 - Updated settings.py for production.

```
• import dj_database_url
• from environ import Env
• env = Env()
• env.read_env()
• ENVIRONMENT = env('ENVIRONMENT', default='production')
• SECRET_KEY = env('SECRET_KEY')
• if ENVIRONMENT == "development":
•     DEBUG = True
• else:
•     DEBUG = False
```

```

•
• ALLOWED_HOSTS = ['edu-learn.up.railway.app']
•
• CSRF_TRUSTED_ORIGINS = ['https://edu-learn.up.railway.app']
•
• INSTALLED_APPS = [
•     'django.contrib.admin',
•     'django.contrib.auth',
•     'django.contrib.contenttypes',
•     'django.contrib.sessions',
•     'django.contrib.messages',
•     'django.contrib.staticfiles',
•     'Course_app',
•     'admin_honeypot',
•     'rest_framework',
• ]
• if ENVIRONMENT == 'development':
•     DATABASES = {
•         'default': {
•             'ENGINE': 'django.db.backends.sqlite3',
•             'NAME': BASE_DIR / 'db.sqlite3',
•         }
•     }
• else:
•     DATABASES = {
•         'default': {
•             'ENGINE': 'django.db.backends.postgresql',
•             'NAME': env('DB_NAME'),
•             'USER': env('DB_USER'),
•             'PASSWORD': env('DB_PASSWORD'),
•             'HOST': 'switchyard.proxy.rlwy.net',
•             'PORT': '36759',
•         }
•     }
• ACCOUNT_USERNAME_BLACKLIST = ['edu_learn']
•
• STATIC_ROOT = os.path.join(BASE_DIR, 'staticfiles')
•
• STATICFILES_STORAGE = 'whitenoise.storage.CompressedManifestStaticFilesStorage'
•

```

- `MIDDLEWARE.insert(1, 'whitenoise.middleware.WhiteNoiseMiddleware')`
-
- `CSRF_COOKIE_SECURE = True`
- `SESSION_COOKIE_SECURE = True`

- Env

```
# Environment
ENVIRONMENT=development

# Django Secret Key (for development use only; change this in production)
SECRET_KEY=django-insecure-5ti_kws7fja&zhc1-$v#hg1uodl-0mtjtmw#gb@z1x_$6h7$6h

# Email (Gmail SMTP)
EMAIL_HOST_USER=shakibrybm@gmail.com
EMAIL_HOST_PASSWORD=egpi ilyt huuw lcux # App password, not your Gmail login password

# PostgreSQL database (used by dj_database_url in production)
#
DATABASE_URL=postgresql://postgres:RBJNrwiBTYOxMVmJOMRRPDYOzQGBlcMs@postgres.railway.internal:5432/railway
DATABASE_URL=postgresql://postgres:RBJNrwiBTYOxMVmJOMRRPDYOzQGBlcMs@switchyard.proxy.rlwy.net:36759/railway

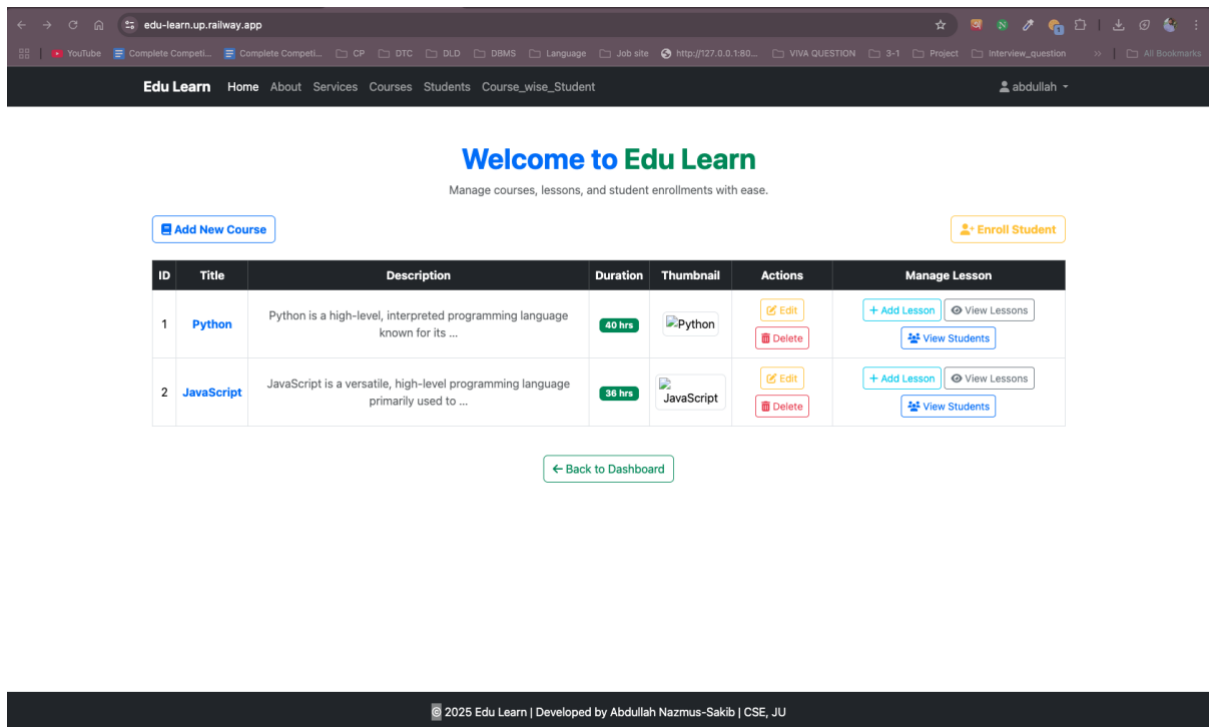
# Optional: Explicit DB info (useful for other tools or debugging, not used by default in Django settings)
DB_NAME=railway
DB_USER=postgres
DB_PASSWORD=RBJNrwiBTYOxMVmJOMRRPDYOzQGBlcMs
DB_HOST=switchyard.proxy.rlwy.net
DB_PORT=36759
```

- requirements.txt.
- `asgiref==3.8.1`
- `dj-database-url==2.3.0`
- `Django==5.2`
- `django-admin-honeypot-updated-2021==1.2.0`
- `django-enviro==0.12.0`
- `django-ipware==7.0.1`

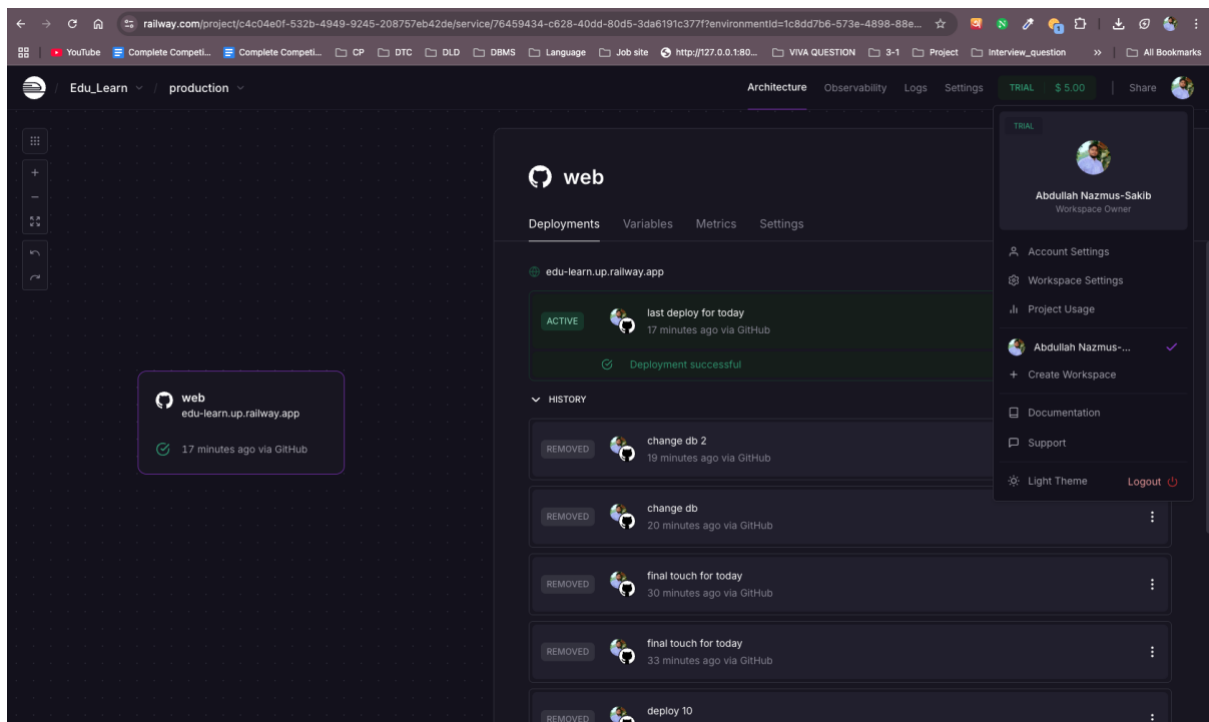
- `django-rest-framework==3.16.0`
- `gunicorn==23.0.0`
- `packaging==24.2`
- `pillow==11.1.0`
- `psycopg2-binary==2.9.10`
- `python-ipware==3.0.0`
- `sqlparse==0.5.3`
- `typing_extensions==4.13.2`
- `whitenoise==6.9.0`
-

- Procfile.
web: gunicorn Edu_Learn.wsgi:application

3. Screenshots of:
 - The live application homepage.



- The admin panel on Railway.



- A successful API response from the live deployment.

edu_learn / Get data

GET

https://edu-learn.up.railway.app/

Send

Params

Authorization

Headers (7)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (1)

Headers (12)

Test Results (1/1)

200 OK

183 ms

6.26 KB

Save Response

HTML

Preview

Visualize

Login

Welcome back! Please login.

Username

Password

Forgot Password?

Login

Don't have an account? [Sign up](#)